Developing and Operationalising Natural Language Processing solutions for Child Safeguarding: A Critical Review of Technical Possibilities and Organisational Realities in UK Policing – TITLE TO BE CONFIRMED!!!

Theoretical Perspectives of Advanced Professional Practice - MOD006047

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Abstract

Purpose: Lorem impsum...

Method: Lorem impsum...

Findings: Lorem impsum...

Conclusion: Lorem impsum...

Contents

1 Introduction **6**
1.1 Research context . . . 6
1.2 Purpose and scope of the paper . . . 7

2 Literature Search Strategy and Methodology **8**
2.1 Review Methodology and Rationale . . . 8
2.2 Tools and software . . . 9
2.3 Search approach and databases . . . 10
2.4 Keywords and search terms . . . 10
2.5 Inclusion and exclusion criteria . . . 15
2.6 Results and filtering process . . . 16

3 The Landscape of Child Sexual Exploitation and Policing's Response **17**
3.1 Scale and trajectory of CSA . . . 17
3.2 The grooming to contact offending pathway . . . 17
3.3 Policing's reactive posture . . . 17
3.4 The case for technological intervention . . . 17

4 NLP and AI in Safeguarding and Criminal Justice Contexts **18**
4.1 NLP applications in public services . . . 18
4.2 Computational approaches to grooming language detection . . . 18
4.3 The data provenance problem . . . 18
4.4 What the technical literature leaves unanswered . . . 18

5 Technology Adoption and Organisational Change in Policing **19**
5.1 Policing as a technology consumer . . . 19
5.2 Tacit versus explicit knowledge cultures . . . 19
5.3 Theoretical frameworks for technology adoption . . . 19
5.4 Structural fragmentation as an implementation barrier . . . 19

6 Synthesis: Where Technical Possibility Meets Organisational Reality **20**
6.1 The convergence problem . . . 20
6.2 Inductive and deductive reasoning across the two tracks . . . 20
6.3 Towards a theoretical position . . . 20

7 Conceptual Framework and Research Questions **21**
7.1 Presenting the conceptual framework . . . 21
7.2 How the literature has shaped the framework . . . 21

7.3 Refined research aim and sub-questions 21

8 Conclusions **22**
8.1 Summary of the argument 22
8.2 Implications for professional practice 22
8.3 Next steps 22

A Patch One - Upload and Feedback **24**

B Patch Two - Upload and Feedback **25**

Acronyms

CEOP Child Exploitation and Online Protection Centre. 15

CSA Child Sexual Abuse. 6–8, 15

ICO Information Commissioner's Office. 15

NIST National Institute of Standards and Technology. 15

NLP Natural Language Processing. 6–8, 15

NPCC National Police Chiefs' Council. 15

PKMS Personal Knowledge Management System. 9

ZK Zettelkasten. 9

List of Figures

2.1 Zettelkasten method application for literature and note management . . . 9

List of Tables

2.1 Search Keywords by Track 10
2.2 Literature Review Search Matrix 12

Chapter 1

Introduction

1.1 Research context

This paper defines the research context before selecting the most appropriate methodology and strategy for the literature review which follows.

Empirical observations, supported by academic and grey literature, (Finkelhor et al., 2024; National Police Chiefs' Council, 2025) identify that reported instances of Child Sexual Abuse (CSA) are increasing, with resultant demand on policing outstripping proactive preventative capabilities.

The gap between demand and capability in both CSA and other operational areas of policing is acknowledged across the sector, generating political and organisational appetite for technological solutions, (National Police Chiefs' Council, 2026). Despite this appetite, evidence from both professional and academic sources identifes that policing is structurally complex and requires reform to better adopt technology, (Cooper, 2024), with cultural barriers further diminishing the successful use of technological solutions, (Thompson and Manning, 2021; Kassem and Erken, 2025).

These challenges are significant, yet they do not negate the potential of emerging technologies to address the identified capability gap. Advances in Natural Language Processing (NLP) technologies present an opportunity to deliver preventative safeguarding in the CSA context, acknowledging that the gap between technical possibility and operational deployment is not primarily technical but organisational.

This paper explores how to bridge this gap, drawing on four literature tracks to develop a conceptual framework capable of addressing the proposed research challenge: *How can natural language processing methods be developed and operationalised to identify precursors of contact sex offending against children, to enhance proactive safeguarding opportunities within the existing operational constraints of policing?*

This challenge consists of four thematic research areas:

1. What linguistic precursors of contact sex offending exist in offender-victim conversations?
2. Which NLP methods can be most effectively realised within policing infrastructure?
3. What implementation barriers exist within policing that may hinder NLP adoption?

4. What strategies can be developed to overcome those barriers?

1.2 Purpose and scope of the paper

This research is inherently interdisciplinary, positioned at the intersection of technical and organisational domains. Accordingly, a literature review will require a multi-track approach, with each track addressing a different aspect of the research challenge. The four tracks are as follows:

1. The first track explores the landscape of CSA and the linguistic characteristics which precede contact offending, establishing the empirical foundation for a technical solution.
2. The second track reviews the technical literature on NLP applications in safeguarding and criminal justice contexts, focusing on grooming language detection and the computational methods available for identifying linguistic precursors of contact offending.
3. The third track examines the organisational dynamics of technology adoption in policing, drawing on both academic literature and empirical evidence of structural and cultural barriers to technical innovation.
4. The fourth track addresses the ethical and governance boundaries within which any NLP solution must operate, encompassing algorithmic bias, data protection, proportionality, and the ethical context surrounding preventative safeguarding.

Each track is explored in depth, and their insights are then synthesised to identify where technical possibilities meet organisational realities. This structure enables a comprehensive understanding of both the capabilities and constraints shaping both the research problem and the practice improvement opportunity and grounding any solution in organisational and ethical realities

Chapter 2

Literature Search Strategy and Methodology

2.1 Review Methodology and Rationale

This literature review adopts a systematic integrative approach. Snyder (2019) documents a number of review types, noting that an integrative review seeks to combine multiple perspectives, synthesise those views, and to develop a new framework or perspective.

The interdisciplinary nature of this research, noted in section 1.2, necessitates a review methodology that can accommodate multiple perspectives and evidence types. An integrative design as defined by Torraco (2016) and Torraco (2005) allows for a review and synthesis across domains, building a conceptual framework which seeks to unify the disparate domains into a framework for operationalising and implementing an NLP solution into a safeguarding CSA context.

A systematic approach is layered into this integrative approach to allow for a targeted and transparent search and selection process in reviewing the evidence, ensuring that the review is both comprehensive and replicable. This systematic layer is particularly important given the breadth of this review, which spans technical, organisational and ethical domains, each with their own bodies of literature and evidence types. Whittemore and Knafl (2005) identifies that by adding a systematic approach to an integrative review, findings and subsequent synthesis across domains can lead to the delivery of evidence based practice improvements.

From an academic perspective, a doctoral literature review needs to be defensible and reproducible. Whilst the integrative approach allows for synthesis and framework development, a layered systematic approach ensures reproducibility whilst adding a counter balance to my own personal biases and my tendency to approach problems with a solutionist mindset.

2.2 Tools and software

The multifaceted nature of this review demands a robust literature and note management strategy supported by appropriate tools. Basu (2020) describes the Zettelkasten (ZK) method which can be used to manage literature sources and create links between them, supporting the synthesis process. This is furthered by Reck (2023) who describes the ZK method as a process of creating notes on individual literature sources, applying tags to those notes and then using the tagging system to create links between the notes. Both Basu (2020) and Reck (2023) describe the ZK method as the process of building a Personal Knowledge Management System (PKMS). Cheong and Tsui (2011) describes a PKMS as a structure which allows for the development of "*personal information management, personal knowledge internalization, personal wisdom creation, and interpersonal knowledge transferring*"

Reck (2023) considers the use of various software tools to support the ZK method. Following a period of experimentation with various applications, a software stack to support the ZK method for this review was selected consisting of Zotero for literature management and bibliography generation, Obsidian for note taking and synthesis and LaTeX for writing and formatting the review.

Zotero was selected for its robust literature management capabilities, including the ability to capture metadata, PDFs and notes on literature sources, and to auto generate and update a master bibliography file. Obsidian was selected for its flexibility in note taking and synthesis, allowing for the creation of a network of linked notes which can be easily navigated and searched. LaTeX was selected for its professional formatting capabilities, particularly for handling citations, figures and tables, and for its ability to draw directly from Zotero's auto-generated bibliography, thus ensuring consistency and ease of referencing.

All content is then automatically synced to a cloud store and defined content saved a second time to a GitHub repository. My use of the ZK method and the associated software stack is illustrated in figure 2.1.

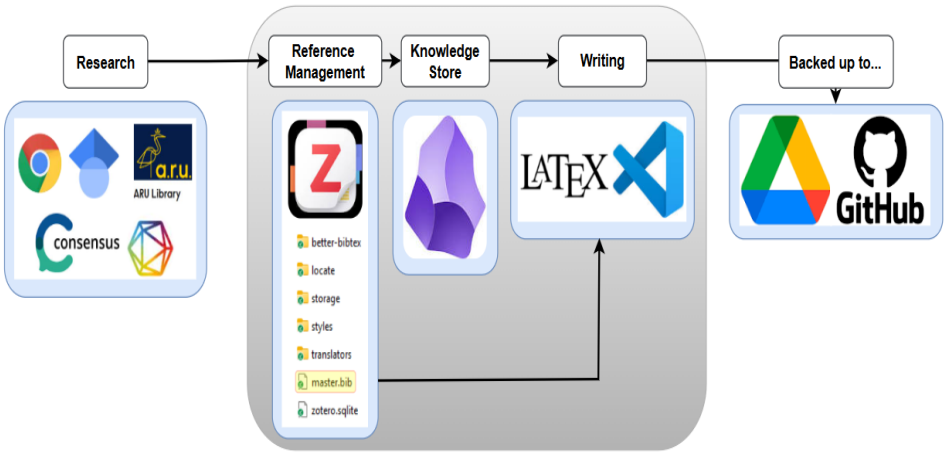

Figure 2.1: Zettelkasten method application for literature and note management

2.3 Search approach and databases

Section 1.2 identifies four distinct research tracks. Each of these tracks has a unique focus and therefore requires a dedicated search strategy to maximise the relevance and quality of included literature. The search strategy presented in table 2.2 defines the four core research tracks, the question to be answered for each track, the selected databases for each track, and the rationale for selecting those databases. The search strategy is designed to ensure that the literature review is comprehensive, targeted and relevant to the research challenge and that it can be replicated by other researchers. The search strategy is also designed to ensure that the literature review is manageable, given the breadth of the research challenge and the interdisciplinary nature of the review.

2.4 Keywords and search terms

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Table 2.1: Search Keywords by Track

Track	Core Terms	Extended & Theoretically-Grounded Terms
Track 1 — Criminology & Behavioural Science		
Criminology & Behavioural Science	child sexual abuse; child sexual abuse material; online grooming; contact offending; offender language; linguistic precursors; grooming typology; offender typology; predatory behaviour; manipulation tactics; forensic linguistics	online-to-offline offending; grooming progression; grooming escalation; grooming stages; coercive communication; trust development; victim selection; sexual communication with a minor; predatory communication; contact offending precursors; CSAM-to-contact trajectory; proactive safeguarding
Track 2 — Computer Science & NLP		
Computer Science & NLP	natural language processing; grooming detection; online predator detection; text classification; transformer models; BERT; large language models; explainability; edge deployment; corpus annotation; predator identification; digital communication analysis	PAN shared task; Perverted Justice corpus; chat log analysis; conversation-level classification; sequential pattern detection; imbalanced classification; real-time text analysis; hate speech detection; toxic language detection; conversational AI detection; forensic linguistics; proactive safeguarding technology
Track 3 — Organisational Studies & Policing		
Organisational Studies & Policing	policing technology adoption; organisational change; law enforcement innovation; institutional barriers; police culture; AI in policing; digital transformation; technology resistance; technology readiness; change management; Diffusion of Innovations; evidence-based policing	tacit knowledge policing; knowledge management policing; police reform; technology scepticism; innovation diffusion public sector; organisational inertia; organisational ambidexterity; change agent policing; technical literacy leadership; national policing technology; fragmented governance; street-level bureaucracy; insider outsider role change
Track 4 — Ethics, Law & Governance		

Continued on next page

Table 2.1 continued

Track	Core Terms	Extended & Theoretically-Grounded Terms
Ethics, Law & Governance	algorithmic bias; AI ethics; data protection; GDPR; proportionality; responsible AI; surveillance governance; AI governance; explainable AI; RIPA; Investigatory Powers Act; Law Enforcement Directive; EU AI Act; HRA Article 8; UK data protection	predictive policing ethics; automated decision-making law enforcement; Data Protection Act 2018; law enforcement processing exemption; data minimisation; purpose limitation; algorithmic accountability; child data protection; misidentification risk; ethical AI deployment; surveillance law UK; rights-based safeguarding; EU-UK regulatory divergence
Cross-Track — Apply in consolidated searches (Scopus; Google Scholar)		
All tracks	proactive safeguarding; preventative intervention; early warning systems; risk prediction; NLP policing; child protection technology; safeguarding automation; AI child safeguarding	

Note. Core terms reflect established disciplinary vocabulary for initial broad searches. Extended terms capture theoretically-grounded, pathway-specific, and dataset-specific language to improve recall in targeted passes. Cross-track terms (proactive safeguarding; preventative intervention; early warning systems; risk prediction) should be applied in consolidated searches across Scopus and Google Scholar.

Table 2.2: Literature Review Search Matrix

Domain	Analytical Question	Databases	Database Justification
Track 1 — Criminology & Behavioural Science			
Criminology & Behavioural Science	*What linguistic and behavioural precursors of contact sex offending exist in offender–victim conversations, and how do grooming typologies characterise the progression toward contact abuse?*	• Scopus • PsycINFO • Web of Science • Criminal Justice Abstracts • NCJRS • CINAHL • ASSIA • LLBA • Google Scholar	• **Scopus** — broad interdisciplinary coverage with strong indexing of criminology and psychology journals. • **PsycINFO** — essential for the psychological and behavioural dimensions of offending, including victimology and typology research. • **Web of Science** — high-quality peer-reviewed social-science journals relevant to behavioural precursors. • **Criminal Justice Abstracts** (via EBSCOhost) — focused criminological literature with strong coverage of sex-offending typologies. • **Google Scholar** — grey literature, NGO reports, and regulatory publications alongside mainstream citation trails.
Track 2 — Computer Science & NLP			
Computer Science & NLP	*Which NLP methods can be most effectively realised within policing infrastructure, and what does the evidence base reveal about model performance, explainability, and operational deployment constraints?*	• IEEE Xplore • ACM Digital Library • ACL Anthology • EMNLP, NAACL & COLING proceedings • Scopus • arXiv (cs.CL, cs.AI, cs.CY) • Semantic Scholar • Papers With Code • LREC-COLING Anthology	• **IEEE Xplore** — essential for applied NLP systems, detection architectures, and deployment pipelines. • **ACM Digital Library** — computational methods, HCI aspects of safeguarding tools, and systems design. • **ACL Anthology** — indispensable for linguistics-grounded NLP, discourse analysis models, and language-detection research. • **arXiv** (cs.CL) — pre-print access to cutting-edge language model research, though quality filtering is required. • **Papers With Code** — essential for model implementations, benchmarks, and reproducibility in detection tasks. • **Scopus** — interdisciplinary bridge between technical and applied criminological literature.
Track 3 — Organisational Studies & Policing			

Continued on next page

Table 2.2 continued

Domain	Analytical Question	Databases	Database Justification
Organisational Studies & Policing	*What structural, cultural, and institutional factors act as barriers to or enablers of NLP adoption in policing organisations, and how does change management theory account for technology resistance in law enforcement?*	• Web of Science • Scopus • Criminal Justice Abstracts • Business Source Complete • PsycINFO • ProQuest • Dimensions • EBSCO Information Science & Technology Abstracts • Google Scholar	• **Web of Science** — high-quality peer-reviewed journals across the social sciences, including organisational behaviour and public administration. • **Criminal Justice Abstracts** — focused coverage of policing and institutional reform literature. • **PsycINFO** — psychological dimensions of organisational resistance and change management in high-accountability environments. • **ProQuest** — strong dissertation access and broad social-science coverage, particularly useful for police reform theses. • **Dimensions** — citation mapping and identification of emerging research clusters in public-sector innovation. • **Google Scholar** — grey literature and policy reports from law-enforcement agencies and inspectorates.
Track 4 — Ethics, Law & Governance			

Continued on next page

Table 2.2 continued

Domain	Analytical Question	Databases	Database Justification
Ethics, Law & Governance	*What ethical and regulatory constraints govern NLP deployment in law enforcement, and how should proportionality, data protection, and rights-based frameworks shape the design and oversight of such systems?*	• Scopus • Web of Science • SSRN • ProQuest • HeinOnline • PhilPapers • EUR-Lex • Dimensions • Google Scholar	• **HeinOnline** — essential for legal scholarship, human rights law, data protection jurisprudence, and regulatory compliance literature. • **SSRN** — pre-print legal scholarship, regulatory theory, and emerging governance frameworks ahead of formal journal publication. • **EUR-Lex** — direct access to EU legislative instruments, including the AI Act and GDPR, relevant to algorithmic oversight. • **PhilPapers** — applied ethics and normative frameworks for algorithmic bias and proportionality in surveillance contexts. • **ProQuest** & **Scopus** — coverage across legal and social-science journals addressing rights-based and regulatory dimensions of AI deployment. • **Dimensions** — citation mapping across interdisciplinary governance and policy research clusters.

Note. Databases assigned per track based on disciplinary indexing coverage. Databases appearing across multiple tracks (Scopus; Google Scholar) should be searched using a consolidated cross-track string to avoid uncoordinated duplication. See Table 2.1 for full keyword sets per track.

2.5 Inclusion and exclusion criteria

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Inclusion Criteria

IC1	Peer-reviewed journal articles, conference proceedings, and edited book chapters published 2018 to present.
IC2	Grey literature from authoritative sources (National Police Chiefs' Council (NPCC), Home Office, Child Exploitation and Online Protection Centre (CEOP), National Institute of Standards and Technology (NIST), Information Commissioner's Office (ICO)) published 2018 to present.
IC3	Studies addressing NLP, machine learning, or computational linguistics applied to safeguarding, criminal justice, or analogous operational contexts.
IC4	Literature examining grooming, CSA, or online predatory behaviour with linguistic, behavioural, or empirical focus.
IC5	Organisational, cultural, or governance studies relating to technology adoption within policing or structurally comparable public sector contexts.
IC6	English-language sources; non-English sources permitted where a peer-reviewed translation or substantive abstract provides sufficient evidential grounding with explicit methodological justification.
IC7	Pre-2018 foundational works where theoretical significance is established and no substantively updated equivalent exists, subject to explicit justification in the synthesis narrative.

– a table showing the inclusion criteria for each track, with a justification for each criterion

Exclusion Criteria

EC1	Sources published prior to 2018 unless meeting the threshold defined in IC7; all exceptions recorded in the review log.
EC2	Non-peer-reviewed opinion pieces, blog posts, or media articles without evidential grounding, unless produced by authoritative professional or statutory bodies.
EC3	Studies focused solely on adult sexual offending with no transferable applicability or analytic relevance to child safeguarding contexts.
EC4	Technical NLP literature without contextual application to safeguarding, criminal justice, or operational deployment scenarios.
EC5	Technology adoption studies from sectors lacking structural or cultural comparability to UK policing (e.g. private sector commercial contexts without analogous accountability frameworks).
EC6	Duplicate publications; where multiple versions of the same study exist, the most recent peer-reviewed version is retained and earlier versions excluded.
EC7	Sources for which full text cannot be obtained through institutional access, interlibrary loan, or open-access repositories within a reasonable retrieval window.

– a table showing the exclusion criteria for each track, with a justification for each criterion

2.6 Results and filtering process

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Chapter 3

The Landscape of Child Sexual Exploitation and Policing's Response

3.1 Scale and trajectory of CSA

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3.2 The grooming to contact offending pathway

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3.3 Policing's reactive posture

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3.4 The case for technological intervention

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Chapter 4

NLP and AI in Safeguarding and Criminal Justice Contexts

4.1 NLP applications in public services

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4.2 Computational approaches to grooming language detection

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4.3 The data provenance problem

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4.4 What the technical literature leaves unanswered

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Chapter 5

Technology Adoption and Organisational Change in Policing

5.1 Policing as a technology consumer

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5.2 Tacit versus explicit knowledge cultures

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5.3 Theoretical frameworks for technology adoption

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5.4 Structural fragmentation as an implementation barrier

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Chapter 6

Synthesis: Where Technical Possibility Meets Organisational Reality

6.1 The convergence problem

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6.2 Inductive and deductive reasoning across the two tracks

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6.3 Towards a theoretical position

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Chapter 7

Conceptual Framework and Research Questions

7.1 Presenting the conceptual framework

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7.2 How the literature has shaped the framework

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7.3 Refined research aim and sub-questions

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Chapter 8

Conclusions

8.1 Summary of the argument

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8.2 Implications for professional practice

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8.3 Next steps

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Appendix A

Patch One - Upload and Feedback

Appendix B

Patch Two - Upload and Feedback

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